

Instance Segmentation of Solar Filaments in GONG $H\alpha$ Images with a Calibrated Ensemble of Detectors

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Abstract—Solar filaments are clouds of cool plasma held above the solar surface by magnetic fields, and their eruptions are a frequent source of coronal mass ejections. Catalogs of filaments are built from full-disk $H\alpha$ images, and segmenting each filament as a separate object in these images is difficult because filaments are thin, fade toward their ends, and are often marked differently by different annotators. This paper describes the system we built for the IEEE BigData Cup 2026 Solar Filament Segmentation Challenge, which uses GONG $H\alpha$ images from the MAGFiLO dataset and scores submissions with panoptic quality (PQ). The system runs four instance segmentation detectors and a semantic segmentation network, redraws every proposed mask with a crop-level U-Net, puts the scores of all models on a common scale with isotonic calibration, and forms each final mask by a weighted vote of the overlapping masks from all models. We describe the design choices, the evaluation procedure we used to accept or reject them, and the results. On held-out validation images, PQ is 0.461 for the full system against 0.450 for the four calibrated detectors alone, and the system scores 0.40 on the public leaderboard. We also report the organizers’ overlap statistics, the coverage of thin filament structures such as barbs, results by observatory, and the agreement between two human annotators on the same images (PQ 0.31–0.34).

Index Terms—solar filaments, H-alpha, GONG, instance segmentation, panoptic quality, ensemble learning

I. INTRODUCTION

Filaments are dense, cool plasma suspended by magnetic fields above polarity inversion lines. In $H\alpha$ images of the full solar disk they appear as dark, elongated structures, and when they erupt they often produce coronal mass ejections and flares. The Global Oscillation Network Group (GONG) [1] records $H\alpha$ images of the full disk around the clock from six observatories, so automatic segmentation of filaments in this stream would make it possible to build filament catalogs and to follow individual filaments over time.

The MAGFiLO dataset [2] provides hand-drawn outlines and spines (center lines) of filaments in GONG images, and the IEEE BigData Cup 2026 Solar Filament Segmentation Challenge uses it to evaluate instance segmentation with panoptic quality (PQ) [3]. Under this metric every filament must be found and outlined as a separate object. Three properties of the data make this hard. Filaments are thin, often only a few pixels wide in a 2048×2048 image, and they fade gradually toward their ends. Neighboring filaments touch or break into pieces. Finally, 42% of the training images were annotated by two or three people, who frequently disagree about which faint structures are filaments and where they end.

Earlier work has applied deep networks to filament segmentation in $H\alpha$ images [4], [5], [6], and Diercke et al. [7] combined an object detector with a segmentation network in a semi-supervised setting. Our system also detects filaments first and segments them afterwards, but it combines several detectors and a semantic segmentation model, since in our experiments no single model came close to the combination.

This paper is an application report. It describes the final system, the experiments that shaped it (including the many alternatives that did not help), and an analysis of the remaining errors. Section II covers the data and the evaluation procedure, Section III the system, Section IV the results, and Section V practical aspects and limitations.

II. DATA AND EVALUATION

Data. The challenge training set contains 707 GONG $H\alpha$ images of 2048×2048 pixels from six observatories, taken between 2011 and 2022. Because some images were annotated by several people, there are 1,154 annotation sets with 8,199 filament outlines in total, each with a spine. The test set has 180 images, and the public leaderboard is computed on about half of them. We used only the training annotations provided by the challenge and no MAGFiLO annotations of test images.

Validation split. We held out 10% of the images (70 images with 116 annotation sets and 908 filaments). The split is by image, so that an image annotated by two people never appears in both training and validation. Each annotation set is scored on its own, as in the challenge. PQ is the sum of the IoUs of matched pairs divided by $|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|$, where a prediction and an annotation match if their IoU exceeds 0.5, and the counts are pooled over all images, following the organizers’ evaluation notebook. We also report its two factors, segmentation quality (SQ, the mean IoU of matched pairs) and recognition quality ($RQ = PQ/SQ$).

Evaluation procedure. Some parts of the system, in particular the score calibration, are fitted on validation data, so scores measured on the same data would be optimistic. For each comparison we therefore split the 70 validation images into two halves, fitted the calibration on one half and scored the other, and repeated this for five random assignments of images to halves, using the same five assignments for every configuration. To judge whether a difference between two configurations is real, we used a paired bootstrap over images with 2,000 resamples. The five splits share the same images

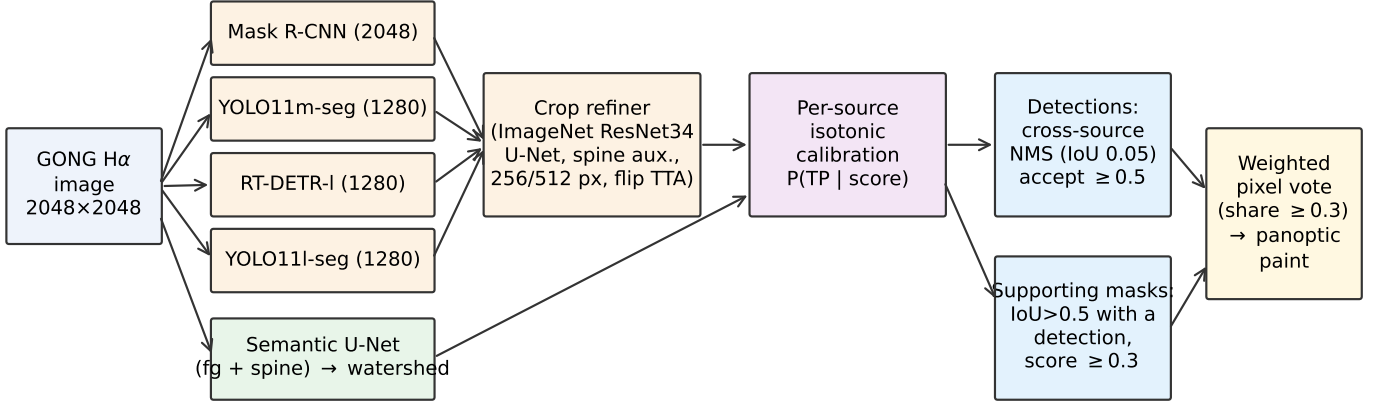


Fig. 1. Overview of the system. Proposals from four detectors are redrawn by a crop refiner, scores are calibrated per model, non-maximum suppression across models selects the detections, and each detection’s final mask is a weighted vote over the overlapping masks from all models. The semantic U-Net contributes both detections and masks.

and the same trained models, so they are not independent. They show how much the result depends on the calibration data, but not how much it would change with a different validation set or retrained models. As a reference, retraining one detector with a different random seed changed PQ by up to 0.005, and we did not treat smaller differences as meaningful. In the later stages of the project we kept a change only if it improved PQ on all five splits.

III. SYSTEM

Fig. 1 gives an overview. The code is available at <https://github.com/biconcavelens/filament-segmentation-2026>. The training settings of all models are listed in Table I.

A. Instance detectors

We use four detectors that differ in architecture and input size: Mask R-CNN [8] (torchvision ResNet-50-FPN v2, pretrained on COCO) at the full 2048-pixel resolution, YOLO11m-seg and YOLO11l-seg [9] at 1280 pixels, and RT-DETR-l [10] at 1280 pixels. In early experiments, many boxes that were placed correctly received very low confidence scores. Increasing the weight of the classification loss reduced this problem (3 times the default for Mask R-CNN, $cls=1.5$ for YOLO and $cls=2.0$ for RT-DETR). For RT-DETR, the median score of boxes overlapping a filament with IoU above 0.5 rose from 0.03 to 0.66. Training Mask R-CNN at the full image resolution rather than at 1333 pixels improved a two-detector version of the system by 0.0035 PQ.

B. Crop refiner

The masks produced by the detectors’ own mask heads are coarse, so we redraw every proposal with a separate network. A square crop 1.8 times the size of the proposal is cut from the full-resolution image, resized to 256×256 pixels, and passed to a U-Net [11] with an ImageNet-pretrained ResNet-34 encoder [12], [13]. Besides the mask, the network also predicts the annotated spine as a second output, which is used

TABLE I
TRAINING SETTINGS (EACH MODEL WAS TRAINED ON A SINGLE 16 GB KAGGLE GPU OR AN 8 GB LAPTOP RTX 4060).

Model	Init., input	Training
Mask R-CNN	COCO, 2048	SGD lr 2.5×10^{-3} , 6 epochs, batch 1, cls. loss $\times 3$, flips
YOLO11m-seg	COCO, 1280	40 epochs, batch 4, $cls=1.5$
YOLO11l-seg	COCO, 1280	103 epochs (time budget), $cls=1.5$
RT-DETR-l	COCO, 1280	60 epochs, batch 2, $cls=2.0$
Crop refiner	ImageNet R34, 256 crop	Adam 3×10^{-4} , cosine, 15 epochs, batch 16, rotation, flips, brightness
Semantic U-Net	ImageNet ConvNeXt-T, 1024 crop	AdamW 3×10^{-4} , cosine, 62 epochs (3 h), batch 4, rotations, flips, photometric

only during training. The loss is BCE plus Dice on the mask (weight 0.7) and BCE on the spine (weight 0.3), and at test time we average the predictions for four flipped versions of the crop. Our first refiner was a U-Net of similar size trained from scratch. Replacing it with the pretrained encoder lowered the validation loss from 0.103 to 0.092 and improved PQ of the four-detector system by 0.006 to 0.012 on all five splits. A second copy of the refiner, trained on 512×512 crops, is used only to provide additional masks for the vote described below.

C. Semantic segmentation model

The fifth model is a U-Net with an ImageNet-pretrained ConvNeXt-Tiny encoder [14] that labels every pixel. It is trained at full resolution on 1024×1024 crops with two outputs, the filament foreground and a 5-pixel-wide drawing of the spines. The foreground is trained with focal loss [15] plus Dice loss [16] and the spine with BCE plus Dice. To turn its output into separate filaments, we threshold the foreground at 0.5, take connected regions of the spine map above 0.3 as seeds, and grow them into the foreground with a watershed, so that touching filaments are split along their spines. Regions

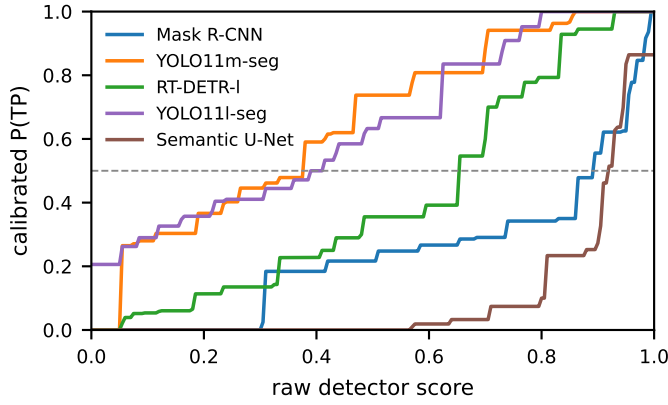


Fig. 2. Isotonic calibration of each model’s score into the probability that a candidate is correct, fitted on validation candidates. The dashed line marks the threshold for accepting a detection.

smaller than 100 pixels are removed, and each filament is scored with its mean foreground probability.

D. Combining the models

The scores produced by different models mean different things (Fig. 2). For example, a candidate has a 50% chance of being a correct detection at a score of 0.38 for YOLO11m-seg, 0.65 for RT-DETR and 0.89 for Mask R-CNN. We therefore map each model’s scores to the probability that a candidate is correct, using isotonic regression [17] fitted on validation candidates. A candidate counts as correct if its redrawn mask overlaps an annotated filament with IoU above 0.5.

Algorithm 1 shows how the candidates are combined in each image. Detections are chosen from the candidates of the four detectors and the semantic model. Candidates are taken in order of decreasing calibrated score, and a candidate is accepted if its score is at least 0.5 and it overlaps no accepted detection by more than IoU 0.05. The mask of each accepted detection is then replaced by a weighted vote over all candidates that describe the same filament. Every remaining candidate with a score of at least 0.3 is assigned to the detection it overlaps most, if that IoU exceeds 0.5, and the final mask keeps the pixels covered by at least 30% of the score-weighted masks in the group. This is similar in spirit to weighted box fusion [18], applied to masks. The masks taking part in the vote come from more sources than the detections. Besides the five sources above, they include the masks from the 512×512 refiner and from running the four detectors on the mirrored image, so that each filament can receive up to 13 masks. These extra masks only change the shape of existing detections and cannot add new ones. Finally, the masks are drawn in order of decreasing score so that no pixel belongs to two filaments, and pieces smaller than 20 pixels are discarded.

IV. RESULTS

A. Overall results

Table II and Fig. 3 show the results of the individual models and of the system as parts are added. On their own, the four

Algorithm 1 Selecting detections and voting on their masks (one image)

Require: candidates (m_i, p_i, s_i) with mask, calibrated score and source; set $\mathcal{D}$ of sources allowed to create detections

- 1: $D \leftarrow \emptyset$
- 2: **for** i in order of decreasing p_i **do**
- 3: **if** $s_i \in \mathcal{D}, p_i \geq 0.5, \text{IoU}(m_i, m_j) \leq 0.05$ for all $j \in D$ **then** $D \leftarrow D \cup \{i\}$
- 4: **end if**
- 5: **end for**
- 6: $G_d \leftarrow \{d\}$ for each $d \in D$
- 7: **for** each candidate $j \notin D$ with $p_j \geq 0.3$ **do**
- 8: $d^* \leftarrow \arg \max_{d \in D} \text{IoU}(m_j, m_d)$
- 9: **if** $\text{IoU}(m_j, m_{d^*}) > 0.5$ **then** $G_{d^*} \leftarrow G_{d^*} \cup \{j\}$
- 10: **end if**
- 11: **end for**
- 12: **for** each $d \in D$ **do**
- 13: $\hat{m}_d(x) \leftarrow [\sum_{k \in G_d} p_k m_k(x) / \sum_{k \in G_d} p_k \geq 0.3]$
- 14: **end for**
- 15: draw $\hat{m}_d$ in order of decreasing p_d onto free pixels; drop pieces < 20 px

TABLE II
VALIDATION RESULTS WITH CALIBRATION FITTED ON HELD-OUT HALVES: MEAN PQ (STANDARD DEVIATION) OVER FIVE RANDOM SPLITS, SQ, RQ, AND THE CHANGE IN PQ RELATIVE TO THE FOUR CALIBRATED DETECTORS (LOWEST AND HIGHEST VALUE OVER THE SPLITS, AND THE NUMBER OF SPLITS WHOSE 95% BOOTSTRAP INTERVAL LIES ABOVE ZERO). ROWS STARTING WITH + ADD TO THE FOUR-DETECTOR SYSTEM. LAST COLUMN: PUBLIC LEADERBOARD.

Configuration	PQ	SQ	RQ	Δ PQ range (sig.)	Pub.
Mask R-CNN	.429 (.004)	.682	.629		
YOLO11m-seg	.437 (.002)	.684	.639		
RT-DETR-l	.436 (.002)	.686	.636		
YOLO11l-seg	.420 (.004)	.687	.611		
Semantic U-Net	.396 (.003)	.669	.591		
4 detectors + NMS	.450 (.003)	.682	.660	–	.40
+ semantic, no vote	.448 (.001)	.678	.661	[−.006, +.003] (0/5)	
+ vote, 4 sources	.453 (.003)	.682	.664	[+.003, +.004] (0/5)	.40
+ vote, 12 sources	.457 (.002)	.684	.668	[+.005, +.009] (3/5)	.40
+ semantic, 5 sources	.455 (.001)	.680	.668	[+.001, +.009] (0/5)	
+ semantic, 13 sources	.461 (.002)	.683	.675	[+.007, +.016] (3/5)	.40

detectors reach PQ 0.42 to 0.44 and the semantic model 0.40. Combining the four calibrated detectors raises PQ to 0.450, mainly because more filaments are found (RQ increases while SQ stays the same). Adding the semantic model’s filaments directly as extra detections does not help, since they bring about as many false detections as correct ones. Voting on the masks raises PQ to 0.457 with 12 mask sources, and with the semantic model added both as a source of detections and as a mask source the full system reaches 0.461. On average it matches 600 of the 908 validation filaments, against 568 for the four detectors. The full system is better than the four detectors on all five splits, and the bootstrap interval lies above zero on three of them. With 70 validation images we consider this consistent but modest evidence.

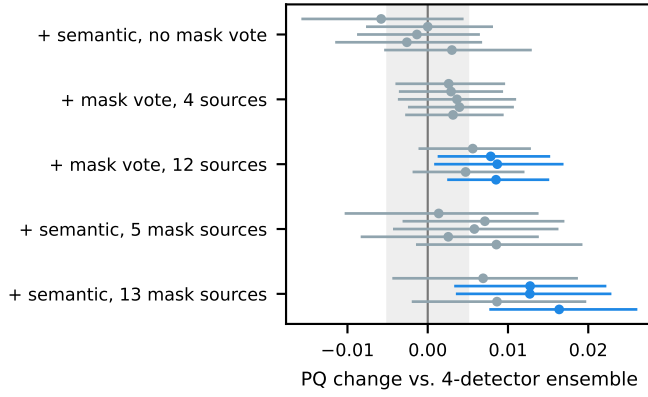


Fig. 3. Change in PQ relative to the four calibrated detectors on each of the five calibration splits (points), with paired-bootstrap 95% intervals. Blue intervals lie above zero. The gray band marks ± 0.005 , the change caused by retraining one detector with a different seed.

TABLE III

COVERAGE OF THIN PARTS (NARROWER THAN ABOUT 5 PIXELS) AND CORE PARTS OF THE ANNOTATED FILAMENTS ON ONE CALIBRATION SPLIT, BY ANY PREDICTION AND BY THE MATCHED PREDICTION OF EACH DETECTED FILAMENT (IOU ABOVE 0.5), AND THE NUMBER OF NEAR MISSES OF THE FOUR-DETECTOR SYSTEM (BEST IOU BETWEEN 0.3 AND 0.5, 96 FILAMENTS) THAT BECAME MATCHES.

Configuration	Any prediction		Matched		Near misses
	Thin	Core	Thin	Core	
4 detectors (256px refiner)	.361	.637	.441	.813	560
4 detectors (512px refiner)	.364	.638	.445	.815	562
+ vote, 4 sources	.395	.656	.478	.832	564
+ vote, 12 sources	.407	.661	.491	.840	569
Full system	.432	.694	.502	.841	600
Semantic U-Net alone	.356	.569	.498	.845	462

On the public leaderboard, a single Mask R-CNN scored 0.27, adding the crop refiner raised this to 0.34, a calibrated combination of two detectors to 0.39, and the pretrained refiner to 0.40. All later versions also scored 0.40. The public score is computed on about 90 images and shown with two decimals, so differences below 0.01 cannot be seen there. Our validation scores were about 0.05 higher than the public scores for every version we submitted.

B. Thin structures and near misses

The organizers name fine structures such as barbs, the thin offshoots along the side of a filament, as one of the main difficulties of the task. Barbs are not labeled separately in MAGFiLO, so we approximated them by the thin parts of each annotated filament. These are the pixels removed by a morphological opening with a disk of radius 2 pixels, that is, parts narrower than about 5 pixels. They include barbs and the narrow ends of filaments and make up 1.9% of the annotated pixels. Table III reports how many of these thin pixels, and of the remaining core pixels, are covered by the predictions.

For the four detectors, 44% of the thin pixels of detected filaments are covered, against 81% of the core pixels. Running the refiner at 512×512 instead of 256×256 hardly changes

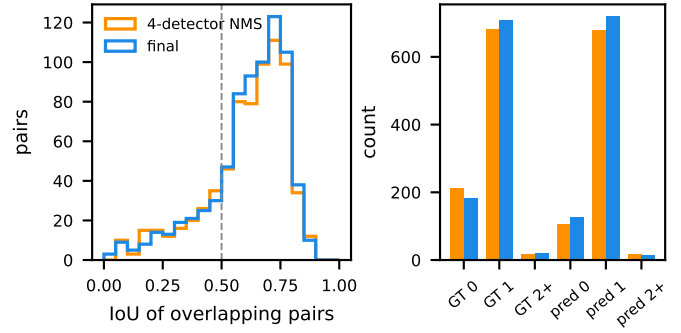


Fig. 4. The organizers’ overlap statistics on one calibration split. Left: IoU of every overlapping pair of prediction and annotation. Right: number of annotated filaments (GT) and predictions that overlap 0, 1, or 2 or more objects of the other kind. “GT 2+” corresponds to fragmentation (one-to-many) and “pred 2+” to merging (many-to-one).

this (44.5%). The mask vote raises thin coverage to 49% and core coverage to 84% without lowering SQ. The gain therefore does not come from simply making masks larger; in an earlier version of the system, enlarging all masks by one pixel lowered PQ from 0.446 to 0.419. The same pattern appears when thin parts are defined with a radius of 3 pixels (coverage 0.58 for the four detectors and 0.63 for the full system). The vote by itself does not create detections. With 12 mask sources it turns 13 of the 96 near misses of the four detectors into matches and loses 4 earlier matches. The 30 additional filaments found by the full system all come from the semantic model; the four detectors had no prediction overlapping them with IoU above 0.3. The semantic model also draws the thin parts of the filaments it finds as completely as the full vote (50%), although on its own it finds far fewer filaments.

C. Organizers’ overlap statistics

The challenge also evaluates the distribution of IoU and Dice over all overlapping pairs of predictions and annotations and the frequency of one-to-many and many-to-one overlaps, computed as in the organizers’ evaluation notebook (Fig. 4). The mean IoU over overlapping pairs is 0.614 for the full system and 0.608 for the four detectors (mean Dice 0.744 and 0.739). The number of annotated filaments with no overlapping prediction falls from 212 to 181 of 908, while the number of predictions that overlap no annotation rises from 104 to 126. Fragmentation and merging are rare in both cases: 19 and 15 annotated filaments overlap two or more predictions, and 14 and 16 predictions overlap two or more annotations.

D. Remaining errors

On one calibration split, the four detectors miss 325 of the 908 validation filaments. For 219 of them, one of the candidates does match the filament, but its calibrated score is below the acceptance threshold (median 0.33). For 67 there is no matching candidate at all, and 57 of these have a candidate with IoU between 0.3 and 0.5, typically a mask that stops before the faint end of the filament. The remaining 39 lose out to an overlapping candidate that is wrong. Lowering the

TABLE IV

PQ BY GONG OBSERVATORY AND BY PERIOD OF SOLAR CYCLE 24 (VALIDATION ANNOTATION SETS, CALIBRATION FITTED ON HELD-OUT HALVES). THE GROUPS ARE SMALL, SO DIFFERENCES OF A FEW HUNDREDTHS SHOULD NOT BE OVER-INTERPRETED.

Group	Sets	Filaments	Per set	4 det.	Full	Δ
Big Bear	21	141	6.7	.411	.438	+.028
Cerro Tololo	21	170	8.1	.445	.468	+.023
El Teide	22	201	9.1	.467	.472	+.005
Learmonth	14	104	7.4	.493	.500	+.007
Mauna Loa	17	124	7.3	.453	.452	-.001
Udaipur	21	168	8.0	.434	.441	+.008
2011–14 (near max.)	74	663	9.0	.447	.462	+.015
2015–18 (declining)	34	220	6.5	.465	.465	+.000
2019–22 (near min.)	8	25	3.1	.385	.404	+.019

acceptance threshold does not help. Accepted candidates with scores between 0.3 and 0.5 are correct at most 32% of the time, even when nine or more sources agree on them, and a detection has to be correct about 34% of the time to increase PQ.

E. Observatories and solar cycle

Table IV breaks down the results by observatory and by period. PQ of the four detectors ranges from 0.41 for Big Bear to 0.49 for Learmonth, and the full system gains most at the two sites where the four detectors do worst. Images taken near the solar minimum contain few filaments (3.1 per annotation set, against 9.0 near the maximum) and have the lowest scores, but there are only 8 such annotation sets.

F. Agreement between annotators

For the 296 training images annotated by two or more people, we scored each annotator’s outlines against another annotator’s with the challenge metric. The resulting PQ is 0.340 (SQ 0.634, RQ 0.536). For the 33 validation images with several annotators the value is 0.309, while our system scores 0.456 against the same annotation sets (0.447 for the four detectors). Because the models are trained on the outlines of all annotators, their output is closer to an average of the annotators than any single annotator is to another. A large part of what PQ counts as error is thus disagreement among annotators about faint structures and filament ends, which limits how high a score any method can reach on this data.

G. Alternatives that did not help

Table V lists the main alternatives we tried and rejected. One pattern stands out. Adding further models as sources of detections, such as a second Mask R-CNN trained with another seed, proposals from mirrored images, or the semantic model without the vote, added false detections faster than correct ones. The same sources did help when they only took part in the mask vote. A gradient-boosted classifier that rescored candidates using agreement between models and shape features ranked candidates better than the calibrated scores but lowered PQ.

TABLE V

ALTERNATIVES THAT WERE TRIED AND REJECTED. CONTEXT: Y = YOLO11M-SEG ALONE; E = FOUR DETECTORS WITH THE EARLIER REFINER TRAINED FROM SCRATCH; F = FOUR DETECTORS WITH THE PRETRAINED REFINER, WITH OR WITHOUT THE MASK VOTE. PQ ON VALIDATION; RANGES OVER THE FIVE CALIBRATION SPLITS WHERE MARKED.

Alternative	Ctx.	Effect on PQ
Mask2Former (Swin-T), frozen or fine-tuned encoder	E	alone $\leq .345$; lowers E
SAM ViT-B decoder with box prompts	–	alone .316
SimSiam pretraining of the Mask R-CNN backbone on 4k unlabeled GONG images	E	no change (.448 vs .449); public .38
Self-training on pseudo-labeled test images	Y	.417 $\rightarrow$.393
CLAHE contrast enhancement	Y	.425 $\rightarrow$.412–.419
Gradient-boosted rescoring of candidates	F	-.014 to -.004 (5 splits)
Growing masks into the U-Net foreground	E / F	+.002 / -.006
Mirrored-image proposals as detections	F	-.009 to +.003 (5 splits)
Second-seed or augmented Mask R-CNN	F	-.005 to +.003 (5 splits)
Averaging two semantic models	F	-.006 to -.003 (5 splits)

V. DISCUSSION

Run time. Most of the inference time goes into redrawing candidates. On average the four detectors produce 93 candidates per validation image above their minimum scores (13, 20, 41 and 19), and each is redrawn with four flipped views. On a 16 GB Kaggle GPU, Mask R-CNN together with the redrawing of its candidates takes 1.9 to 2.2 s per image, and the semantic model 1.2 s per image at full resolution with four flipped views. The full system redraws every candidate three times (256-pixel crops, 512-pixel crops and the mirrored image). A lighter version that uses only the five detection sources in the vote keeps about 45% of the improvement (0.005 PQ) with about a third of the redrawing cost, which may be preferable when processing the full GONG archive.

Limitations. The acceptance and suppression thresholds and the vote parameters were chosen on the same 70 validation images on which we report results. For the vote parameters this matters little, since PQ changed by only 0.001 over a grid of 27 settings, but the other settings may be slightly optimistic. All models were trained once on one split, so our evaluation does not capture the variation from retraining. Thin structures remain the weakest point: even the full system covers only about half of the thin pixels of the filaments it finds, against 84% of the core. Neither the refiner nor the semantic model is trained to give thin structures extra weight, and barbs are also drawn inconsistently by the annotators. Finally, the 0.05 gap between our validation and public scores was stable across versions, and we have not found its cause.

Future work. For thin structures, the semantic model already draws them as completely as the full vote, so a full-resolution model trained with extra weight on thin parts is a

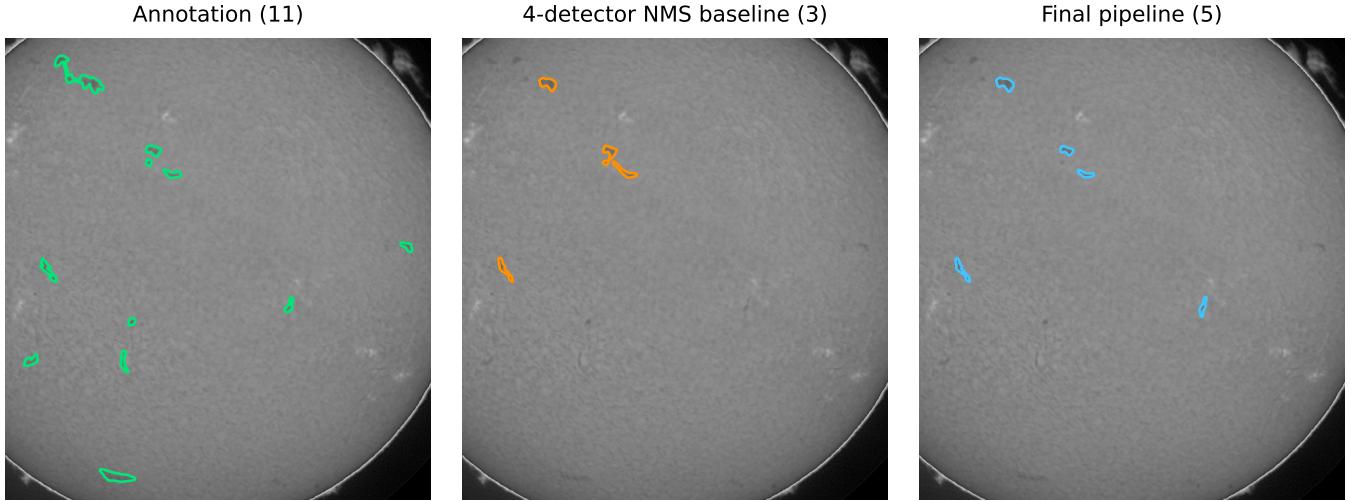


Fig. 5. A validation image on which the mask vote helps (chosen as the image with the largest gain in matched filaments, so it is not a typical case). Left: the outlines of one annotator. Middle: the four detectors merge two filaments into one object and miss several faint ones. Right: the full system separates the two filaments and finds two more, but several faint annotated filaments are still missed.

natural next step. A topology-preserving loss (cIDice) in the crop refiner did not help in our experiments. For recall, most missed filaments have a matching candidate with too low a score, and information at the scale of the whole image, such as co-temporal magnetograms that show the polarity inversion lines along which filaments lie, could help to separate them when used as an additional training signal. Sequences of unlabeled GONG images could also be used for semi-supervised training with consistency between neighboring frames; self-supervised pretraining on single unlabeled frames gave no improvement in our tests.

VI. CONCLUSION

We described a system for instance segmentation of solar filaments in GONG $H\alpha$ images, built for the IEEE BigData Cup 2026. It combines four instance detectors and a semantic segmentation model, redraws every candidate with a pretrained crop refiner, calibrates the scores of all models, and forms each final mask by a weighted vote of the overlapping masks. On held-out validation images it reaches PQ 0.461, compared with 0.450 for the four calibrated detectors alone and 0.42 to 0.44 for the individual detectors, and it scores 0.40 on the public leaderboard without using test annotations. The mask vote improves the outlines of detected filaments, including their thin parts, and the semantic model adds filaments that the detectors miss. Thin structures and faint filament ends remain the main source of error, and the low agreement between human annotators (PQ 0.31 to 0.34) should be kept in mind when comparing results on this benchmark.

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to implement and run experiments and to draft text; all design decisions, results and claims were reviewed by the author.

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